

ALTENRHEIN, Switzerland

WMO: 066900

Lat: 47.4836N

Lon: 9.5667E

Elev: 1307

StdP: 14.01

Time Zone: 1.00 (EUC)

Period: 10-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	16.9	21.6	8.9	9.1	17.3	15.7	12.8	24.7	22.4	53.2	18.1	43.4	6.4	100	0.460

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.6	85.3	69.1	82.2	67.5	79.0	66.3	70.9	81.8	69.2	79.2	67.6	76.8	5.4	280

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
67.1	104.7	77.1	65.6	99.2	75.0	64.3	94.6	73.0	35.5	82.4	34.0	79.3	32.8	76.9	78.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
17.6	14.2	11.6	DB	12.3	90.8	4.9	3.7	8.8	93.4	5.9	95.6	3.2	97.6	-0.4	100.3
			WB	11.5	74.6	5.2	2.1	7.8	76.1	4.7	77.4	1.8	78.5	-2.0	80.0

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	51.2	35.4	35.1	43.3	50.6	57.1	65.0	67.9	67.3	59.8	51.7	43.0	36.9	(6)
	DBStd	13.30	7.01	7.73	5.77	6.64	6.29	5.72	5.20	5.24	5.09	5.79	6.09	6.37	(7)
	HDD50	1851	453	417	215	71	12	0	0	0	0	50	225	408	(8)
	HDD65	5362	917	837	672	433	255	70	30	36	170	411	660	871	(9)
	CDD50	2287	1	0	8	90	231	451	554	537	295	105	14	1	(10)
	CDD65	323	0	0	0	2	9	71	119	108	14	0	0	0	(11)
	CDH74	2196	0	0	0	23	83	531	836	665	56	2	0	0	(12)
	CDH80	517	0	0	0	3	7	136	217	153	1	0	0	0	(13)
(14) Wind	WSAvg	4.4	4.5	4.6	4.9	5.0	4.7	4.6	4.4	4.1	3.8	3.9	4.1	4.5	(14)
(15) Precipitation	PrecAvg	55.1	2.7	3.0	3.5	4.0	5.4	6.7	6.6	6.7	4.9	3.8	4.1	3.6	(15)
	PrecMax	70.8	5.9	8.3	7.0	7.4	12.6	12.4	12.2	11.5	7.9	8.2	9.4	7.6	(16)
	PrecMin	42.0	0.3	0.8	1.2	1.3	0.7	2.6	2.6	1.9	2.2	0.4	0.3	0.4	(17)
	PrecStd	6.1	1.6	1.6	1.5	1.7	2.6	2.4	2.6	2.4	1.5	1.7	2.0	1.7	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	57.3	58.8	64.9	77.1	80.7	87.6	90.1	88.0	78.8	72.0	67.5	62.8	(19)
		MCWB	49.3	46.1	50.6	57.0	63.0	69.9	70.7	70.1	66.5	55.6	51.2	46.8	(20)
	2%	DB	50.3	52.4	61.0	71.3	75.8	83.9	85.5	84.2	75.4	66.2	60.6	53.4	(21)
		MCWB	44.8	45.2	48.5	54.8	60.5	67.9	68.5	69.5	64.9	58.7	52.5	46.9	(22)
	5%	DB	46.7	48.1	57.6	68.0	72.7	80.3	82.0	80.7	71.9	62.8	54.8	48.9	(23)
		MCWB	42.9	42.7	47.7	53.9	59.1	66.7	67.4	68.3	63.5	57.5	49.9	44.4	(24)
	10%	DB	44.2	44.6	53.7	63.3	69.4	76.2	78.6	77.4	68.9	59.9	51.4	46.0	(25)
		MCWB	41.1	40.6	46.2	52.2	58.0	64.7	65.9	66.9	62.0	55.8	47.9	42.5	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	50.0	49.4	52.5	59.1	65.7	73.8	72.8	72.8	68.0	61.0	54.6	50.1	(27)
		MCDB	57.2	56.6	62.3	72.4	76.5	85.7	85.3	83.3	75.7	66.2	60.4	58.3	(28)
	2%	WB	46.0	45.5	50.3	56.7	62.9	70.0	70.4	70.9	66.0	59.2	52.4	47.7	(29)
		MCDB	49.4	51.3	59.0	68.0	72.4	81.1	81.7	81.3	73.5	63.9	58.4	52.7	(30)
	5%	WB	43.3	43.0	48.4	55.1	60.9	67.7	68.5	69.3	64.4	57.8	50.3	45.2	(31)
		MCDB	46.1	47.3	55.4	65.6	69.5	77.8	79.0	78.7	71.4	61.8	55.4	48.8	(32)
	10%	WB	41.3	40.7	46.7	53.2	58.8	65.8	67.0	67.8	62.5	56.3	48.4	42.9	(33)
		MCDB	43.6	44.3	53.2	62.0	66.7	74.6	76.6	76.1	68.0	59.7	51.4	45.4	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	9.4	11.2	16.0	17.4	16.6	16.2	16.6	16.6	14.8	12.4	9.8	9.7	(35)
		MCDBR	13.8	18.7	24.3	25.5	23.1	23.3	23.3	21.9	18.7	16.9	17.0	15.1	(36)
	5% WB	MCWBR	10.0	11.5	14.0	12.8	10.9	10.9	10.3	10.3	10.1	10.2	10.1	10.0	(37)
		MCWBR	13.3	17.9	20.7	23.5	20.5	21.0	21.2	20.4	18.0	15.4	15.6	14.4	(38)
(40) Clear-Sky Solar Irradiance	taub		0.291	0.317	0.358	0.399	0.414	0.420	0.422	0.413	0.383	0.351	0.318	0.287	(40)
		taud	2.410	2.336	2.307	2.241	2.251	2.276	2.268	2.308	2.371	2.446	2.476	2.429	(41)
	Ebn at Noon		252	267	272	270	270	268	266	263	261	252	240	239	(42)
		Edh at Noon	22	29	35	41	42	41	41	38	32	26	20	19	(43)
(44) All-Sky Solar Radiation	RadAvg		366	633	1013	1371	1581	1809	1750	1530	1144	722	401	301	(44)
	RadStd		40	86	134	194	161	170	161	159	119	78	49	46	(45)

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (111 neighbors)		N/A	N/A	N/A	N/A	+0.79	+0.67	N/A	N/A	+95	+25	

Nomenclature: See separate page